

AI & DATA SCIENCE FINAL PROJECT

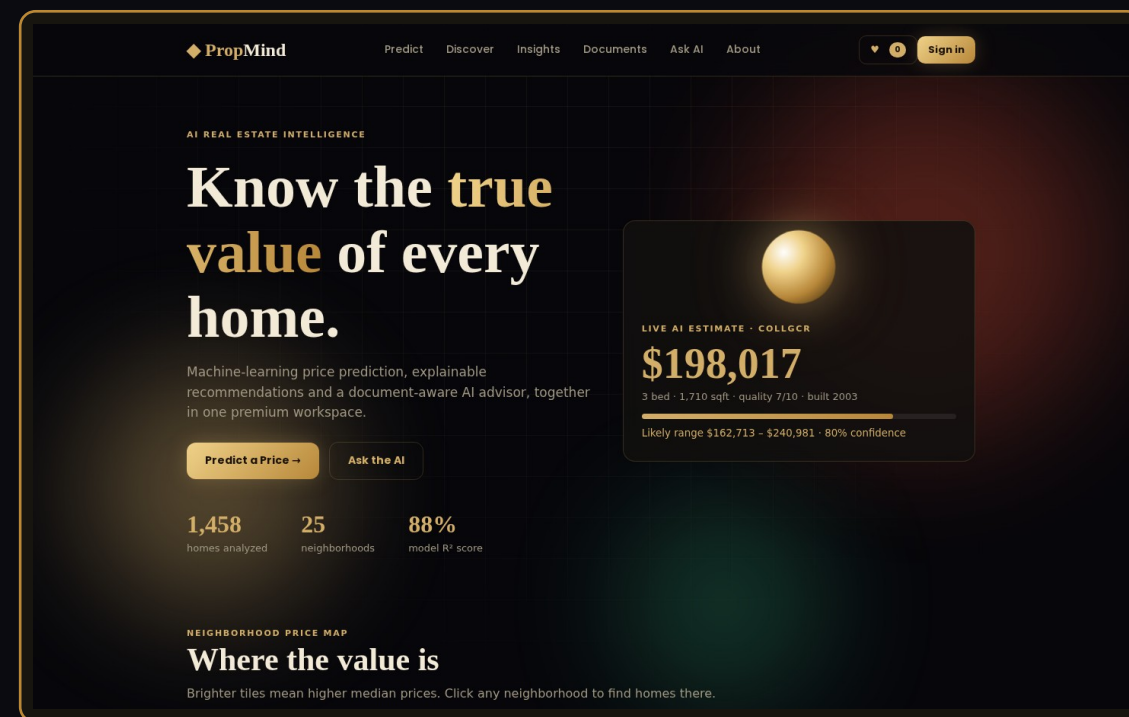
◆ PropMind

AI Real Estate Intelligence Platform

Predict prices · Explained recommendations · AI advisor ·
Document RAG

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SMIT (Saylani Mass IT Training) · October 2026



Buying a home is a decision made in the dark



Opaque pricing

Is this asking price fair for this specific home?



Manual matching

Filtering is slow and nobody explains why a home fits.



Dense documents

Guides and legal steps are long and difficult to search.



Fragmented tools

Valuation, search, advice and documents sit in different places.

PropMind turns these four gaps into one explainable product.

One platform, five capabilities



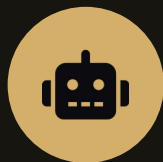
Predict

Fair price, likely range
and what-if effects



Discover

Ranked homes, each
with reasons



AI advisor

Prompt-engineered
Groq assistant



Document RAG

Answers with
citations from
documents

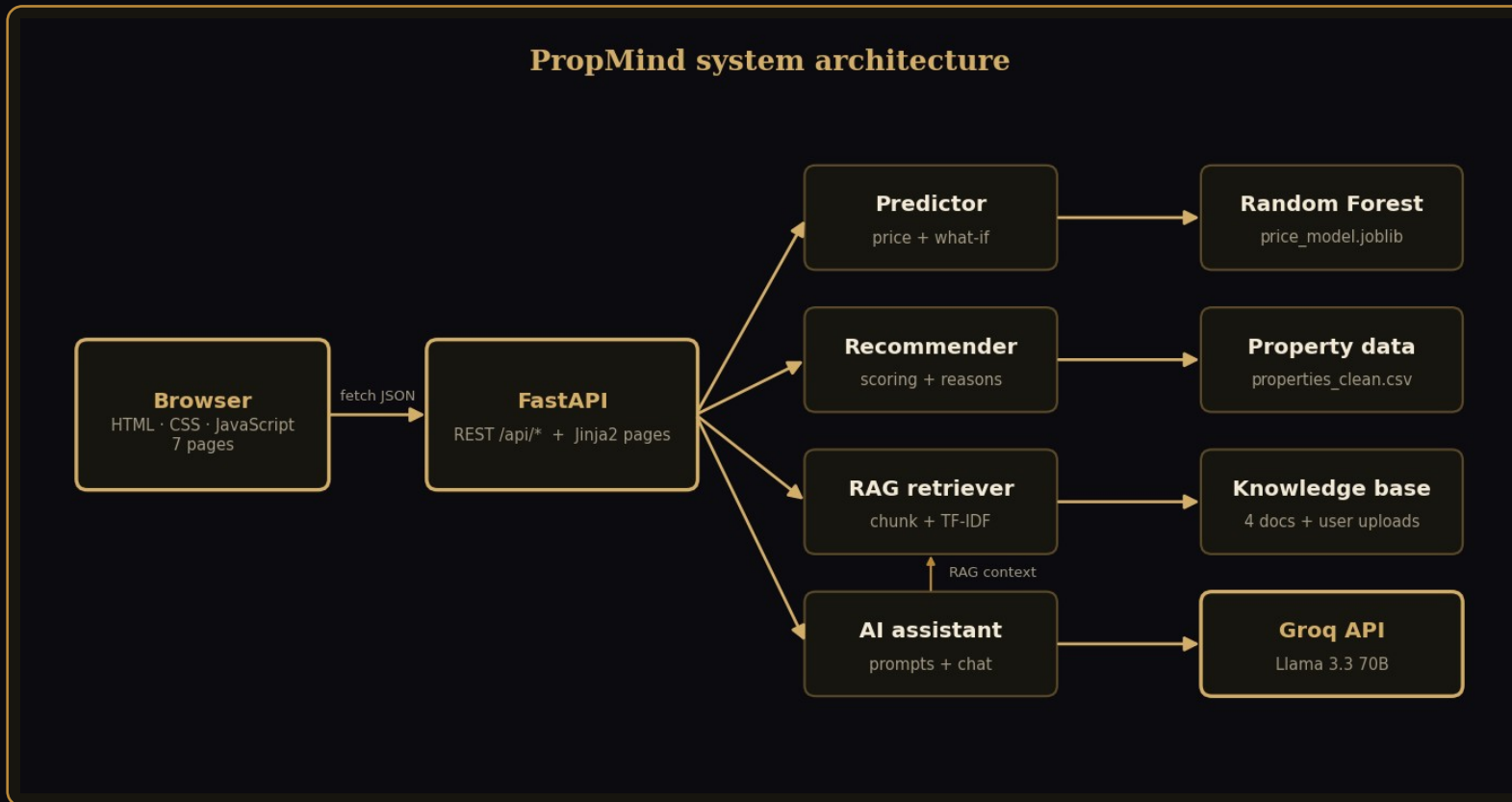


Insights

Neighborhood and
model analytics

Python · scikit-learn · FastAPI · Groq (Llama 3.3) · TF-IDF RAG · HTML, CSS, JavaScript

A clean architecture: thin API, testable services



Thin API layer

FastAPI serves JSON and the seven pages.

Testable services

Plain Python modules with 10 automated tests.

Graceful AI

Groq answers, with an offline fallback.

Data: Ames housing sales, cleaned and understood

1,458

cleaned home sales, no missing values

25

Ames neighborhoods

79 → 9

features in the data vs. model inputs

Cleaning highlights

- NA often means the feature is absent, so it became its own category
- Lot frontage filled from the neighborhood median
- Two known outliers removed (huge area, very low price)
- New features: TotalSF, HouseAge and PricePerSqft

Location and quality drive the price



3.6×

median price gap: NridgHt vs. MeadowV

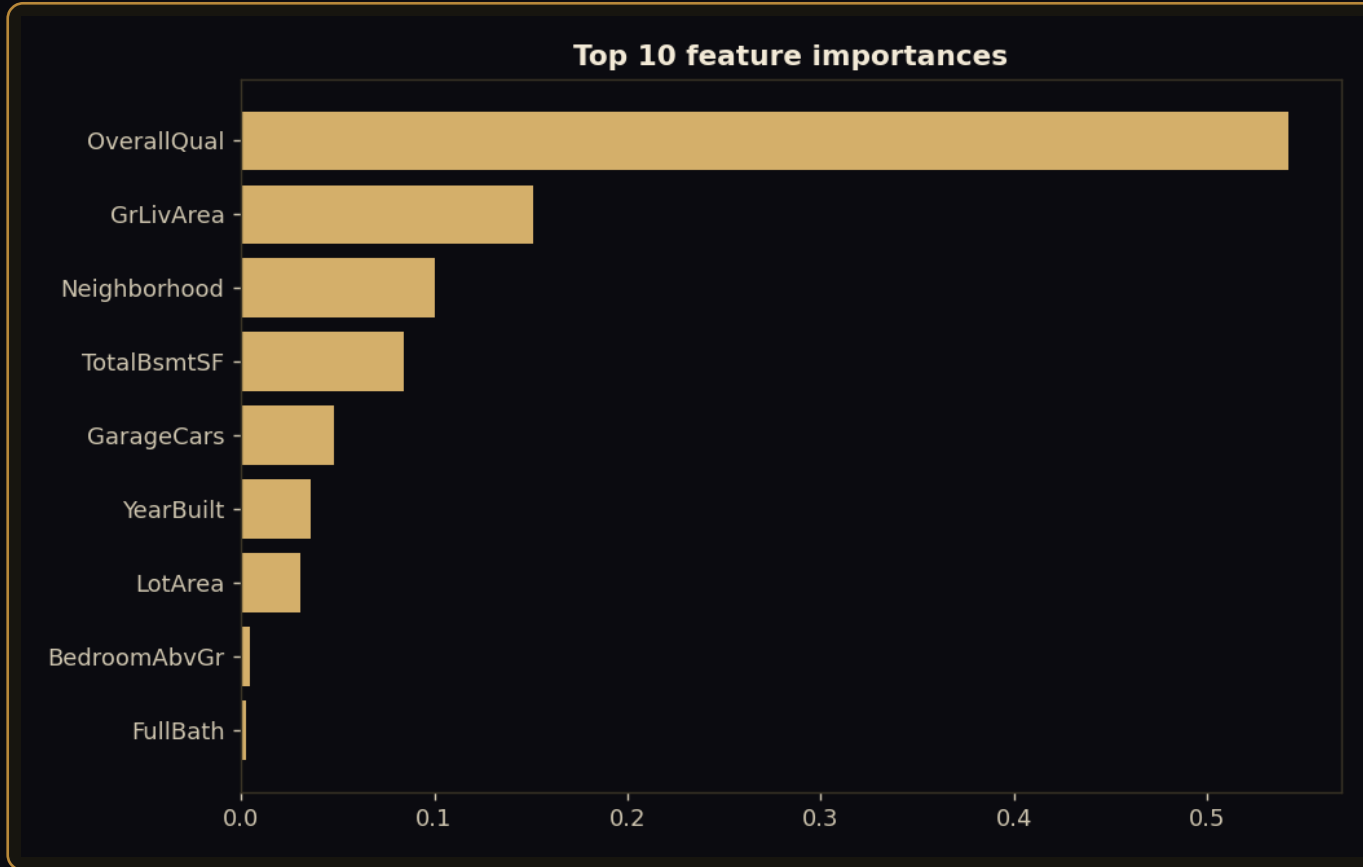
0.80

correlation of quality with price

1.88 → 0.12

price skewness before and after log

Overall quality is the strongest signal



54%

of the model's importance is overall quality

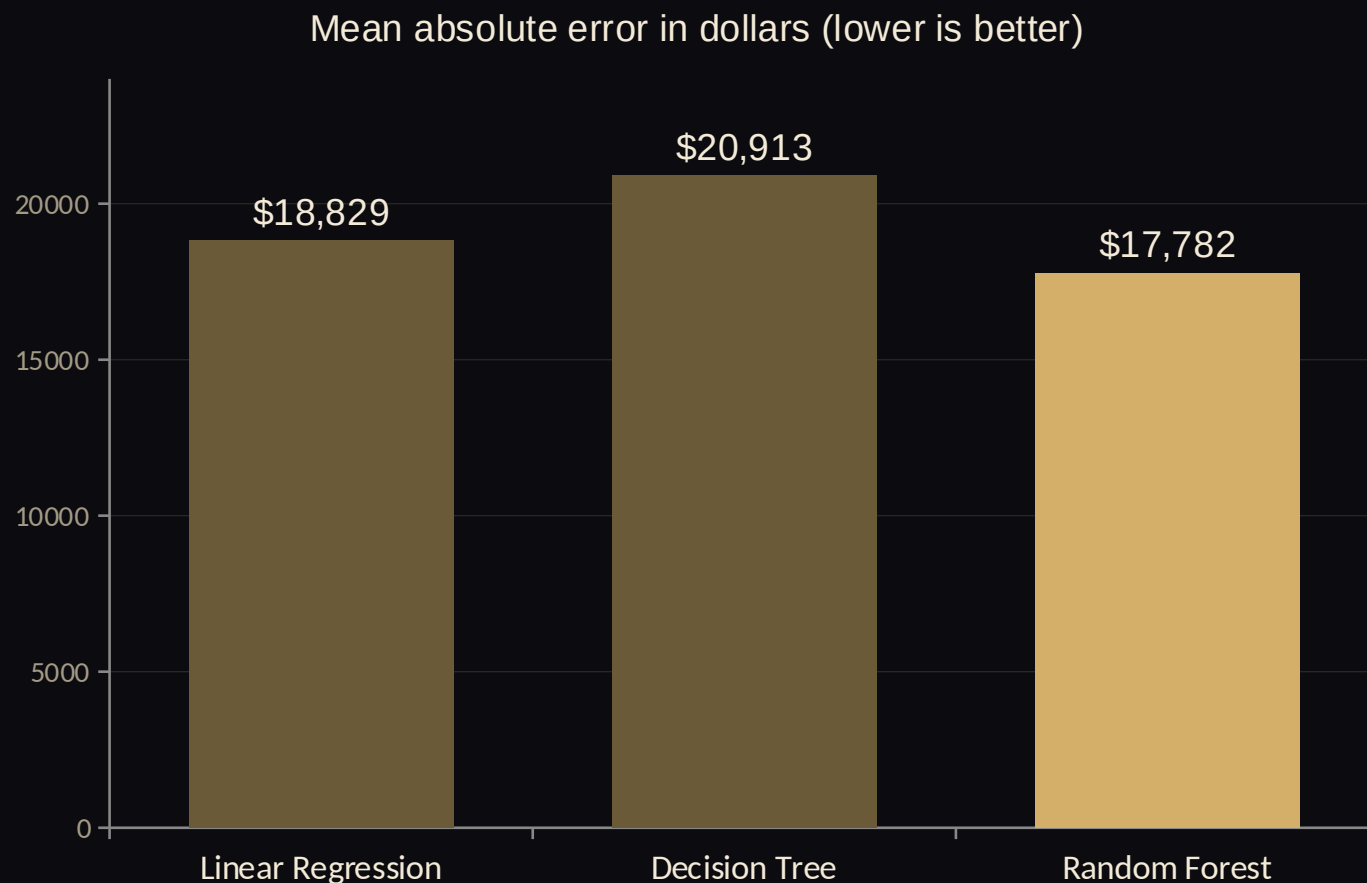
Living area 15%

Neighborhood 10%

Basement size 8.5%

Garage capacity 4.8%

Three models compared, Random Forest deployed



0.885

R^2 of the deployed Random Forest

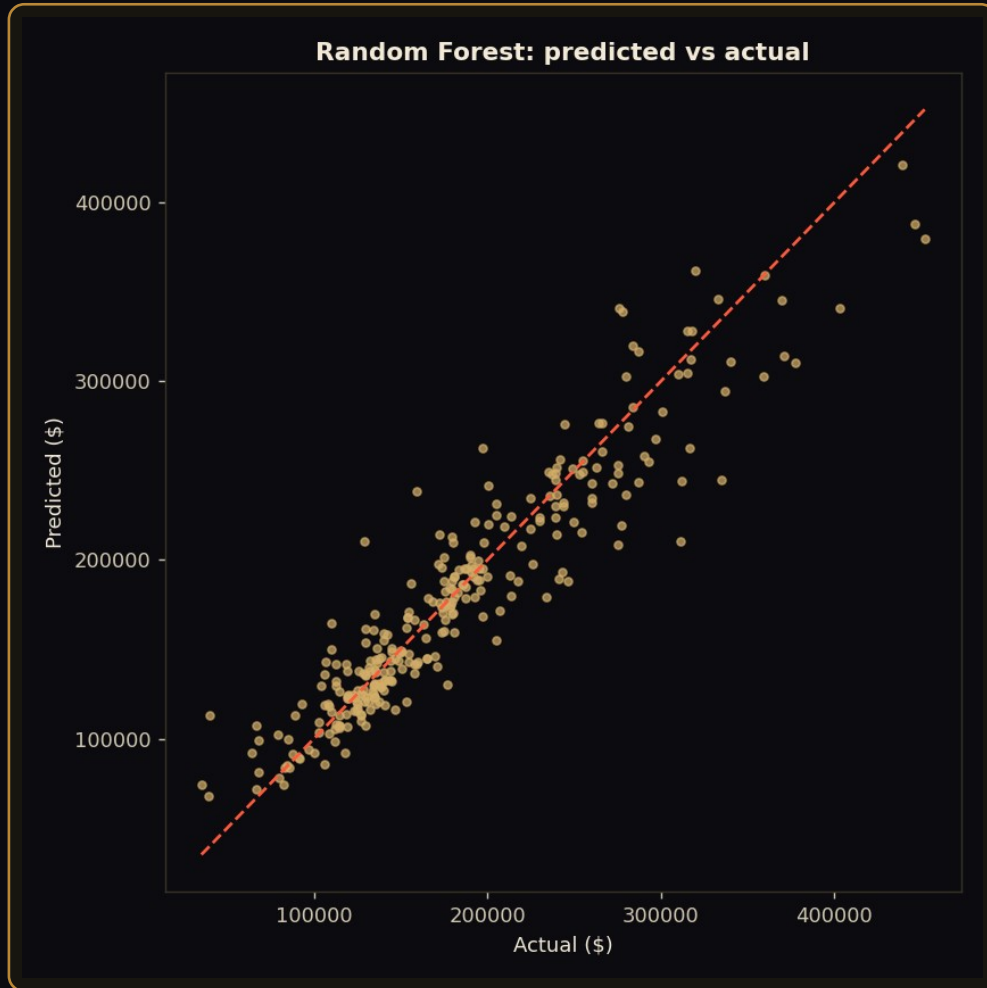
0.876

R^2 of Linear Regression, close behind

0.840

R^2 of the single Decision Tree

How close are the predictions?



63%

of test homes within 10% of the
real price

77%

within 15% of the real price

87%

within 20% of the real price

6.6%

median absolute percentage
error

A bug that only testing could find

BEFORE: one-hot encoding

\$203,204

the same estimate for most neighborhoods, so choosing a location changed nothing

AFTER: target encoding

\$165K – \$211K

range for the same home across neighborhoods, with 21 distinct values

Test the model as a product, not only with a metric.

Recommendations that explain themselves

PropMind Predict Discover Insights Documents Ask AI About

RECOMMENDATION ENGINE

Discover Homes

Tell us your budget and preferences. Every match comes with an explanation of why it was recommended.

Budget \$250,000

Minimum bedrooms 1+ 2+ 3+ 4+ 5+

Preferred neighborhoods (1) Blmngtn Blueste BrDale BrkSide ClearCr CollgCr Crawfor Edwards Gilbert IDOTRR MeadowV Mitchel

Preferences 2+ car garage Newer build (2000+) High quality Large lot Spacious Big basement

Find homes

Recommendations Saved (2) 108 homes fit · showing top 6

#1 MATCH · COLLGCR

\$245,350

8.6% below AI value

3 bed 2 bath 1,694 sqft Quality 8/10 Built 2008 3 car

AI fair value \$268,289 · \$145/sqft

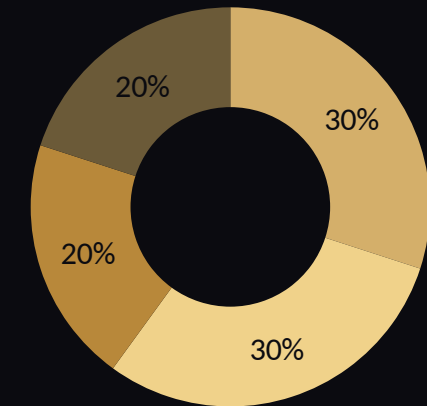
- ✓ Priced at \$245,350, \$4,650 under your \$250,000 budget
- ✓ Listed 9% below the AI fair-value estimate of \$268,289
- ✓ 3 bedrooms, meeting your 3+ requirement
- ✓ Located in CollgCr, your preferred area
- ✓ Matches your preference: has a 3-car garage

This 3-bedroom home in CollgCr offers 1,694 sqft, was built in 2008 and has a quality rating of 8/10. It is priced at \$245,350, about 9% below the AI fair-value estimate of \$268,289. Verify condition with an inspection before making an offer.

Offline summary

2 of 3 selected Compare Clear

Match score weights

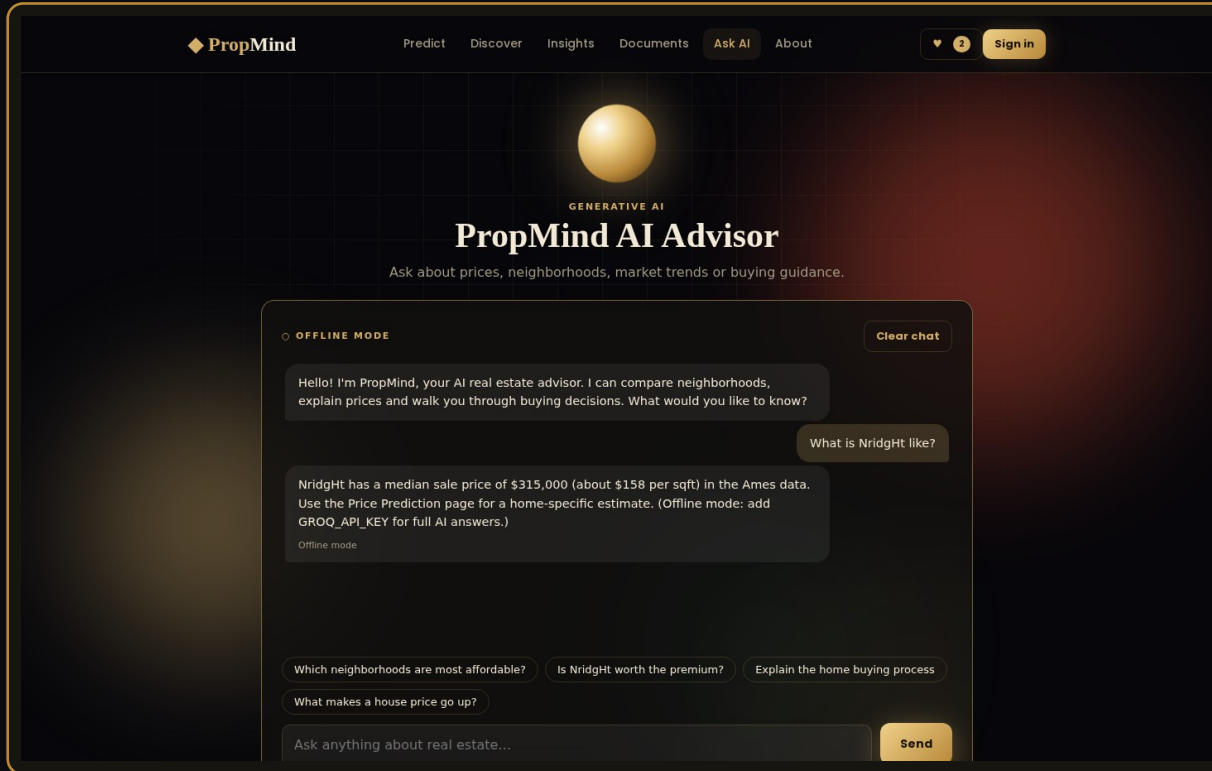


■ Budget fit ■ Value vs AI price
■ Quality ■ Preferences

"Listed 9% below the AI fair-value estimate of \$268,289"

An example of a real reason shown with a result

An AI advisor with guardrails



Role and tone

A concise professional advisor



Real market facts

Statistics injected from the data



No invented numbers

Figures only from context or the user



Professional limits

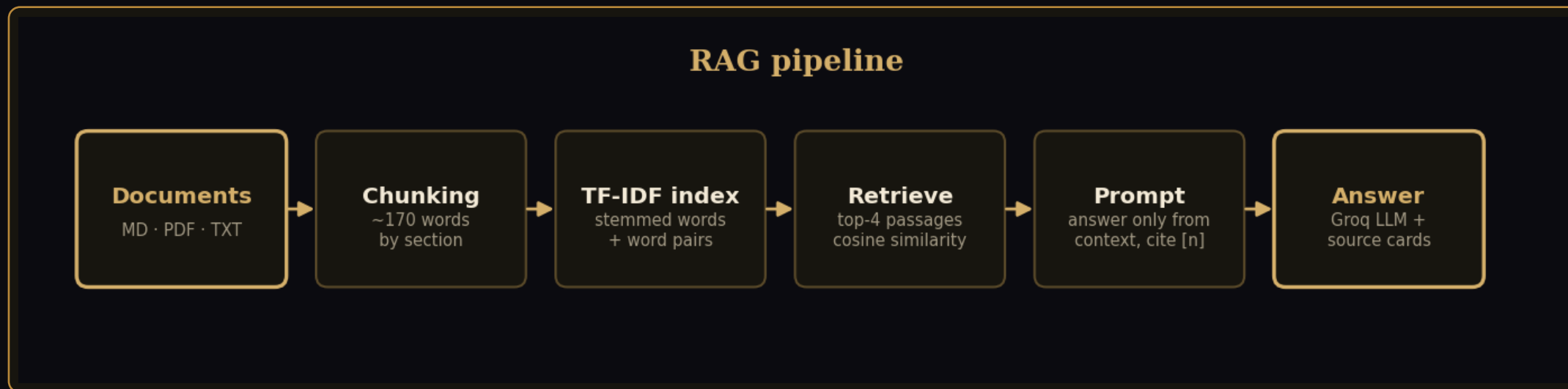
Not legal, tax or financial advice



Offline fallback

Keeps working without an API key

RAG: answers grounded in real documents



4 docs

buying guide, Iowa legal process, investing, area data

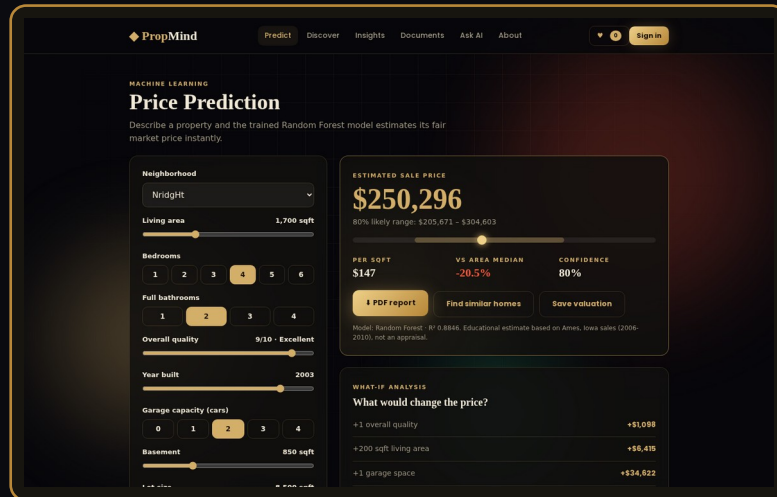
100%

right document ranked first on 15 questions (baseline 93%)

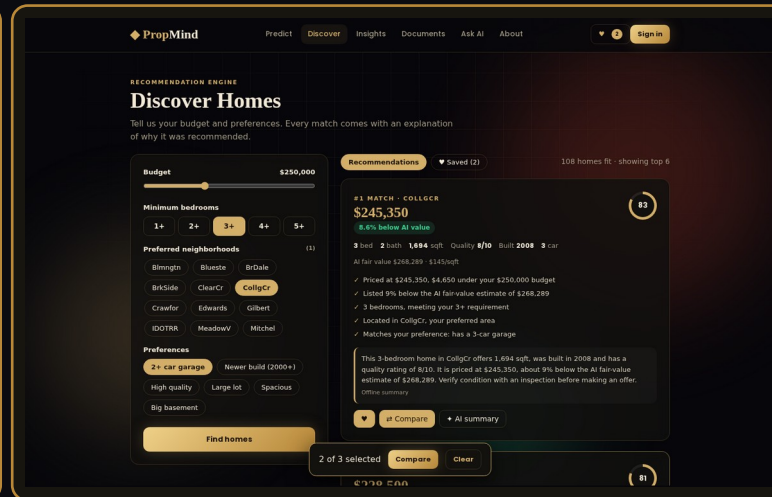
[1] [2]

every answer cites its source passages

Product tour



Price prediction with what-if analysis



Recommendations with reasons and compare



Interactive market insights

Also: login, saved homes, compare up to three, AI summaries, PDF report, document upload

Built like a real product



10 automated tests

prediction, recommender, RAG and fallbacks



GitHub

source, data, tests



Browser-tested pages

filters, compare, PDF, chat and sign-in



Vercel

backend + website



Offline-safe

works without an API key



Firebase

static frontend

Limitations and what comes next

Limitations

- One city, 2006-2010: educational, not an appraisal
- Nine inputs ignore renovations and exact location
- TF-IDF matches words, not meaning
- Demo login lives only in the browser

Next steps

- Embedding-based semantic retrieval
- Real accounts with Firebase Authentication
- Gradient boosting and a larger, newer dataset
- Map view and listing photos
- Urdu and English interface

Estimates are educational, not appraisals: the honest limits are part of the design.

Thank you

Questions?

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Anushay Ayat · AI & Data Science · SMIT